

WEEK 5 PROJECT REPORT

HEALTHCARE DATA CLEANING & DATA MODELING

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Project Title: Healthcare Data Analysis using Power BI

Tool Used: Microsoft Power BI Desktop

Dataset: Healthcare Dataset

Introduction

The Week 5 Power BI project focuses on **Healthcare Data Cleaning and Data Modeling**. The objective of this project was to prepare healthcare patient records for analysis by cleaning the raw dataset, creating meaningful categories, organizing the data into dimension and fact tables, establishing relationships, and creating DAX measures for key healthcare metrics.

The project uses the **Healthcare Dataset** containing information related to patients, admissions, medical conditions, hospitals, doctors, billing, insurance providers, medications and other healthcare-related attributes.

Problem Statement

A healthcare organization needs to analyze patient information, medical conditions, admissions and healthcare-related characteristics.

The raw healthcare data needs to be **cleaned, standardized, transformed and properly modeled** before it can be used to create an interactive Power BI report.

The project therefore focuses on creating a structured data model that can support healthcare analysis and reporting.

Objectives

The main objectives of Week 5 were:

- Import and inspect the healthcare dataset in Power BI.
- Clean and transform the healthcare data using Power Query.
- Handle missing or null values where required.
- Standardize categorical fields such as admission types.
- Create meaningful derived columns such as **Age Group**.
- Create dimension tables and a fact table.
- Build a **star-schema data model**.
- Establish relationships between patient/admission dimensions and billing facts.

- Create DAX measures for important healthcare KPIs.
- Prepare the cleaned data model for healthcare analysis.

Data Preparation and Cleaning

The healthcare dataset was imported into Power BI and inspected using Power Query.

The main dataset used in the project was: **healthcare_dataset**

healthcare_dataset.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance
Bobby JacksOn	30	Male	B-	Cancer	31-01-2024	Matthew Smith	Sons and Miller	Blue Cross
Leslie TERry	62	Male	A+	Obesity	20-08-2019	Samantha Davies	Kim Inc	MediCare
Danny sMitH	76	Female	A-	Obesity	22-09-2022	Tiffany Mitchell	Cook PLC	Aetna
andrew waTtS	28	Female	O+	Diabetes	18-11-2020	Kevin Wells	Hernandez Rogers and Vang	Medicaid
adRIENNE beIl	43	Female	AB+	Cancer	19-09-2022	Kathleen Hanna	White-White	Aetna
EMILY JOHNSOn	36	Male	A+	Asthma	20-12-2023	Taylor Newton	Nunez-Humphrey	UnitedHealth
edwArD EDWaRDs	21	Female	AB-	Diabetes	03-11-2020	Kelly Olson	Group Middleton	Medicaid
CHrIStinA MARTinez	20	Female	A+	Cancer	28-12-2021	Suzanne Thomas	Powell Robinson and Valdez	Cigna
JASmiNe aGullaR	82	Male	AB+	Asthma	01-07-2020	Daniel Ferguson	Sons Rich and	Cigna
ChRISTopher BerG	58	Female	AB-	Cancer	23-05-2021	Heather Day	Padilla-Walker	UnitedHealth
mlchELLE daniELs	72	Male	O+	Cancer	19-04-2020	John Duncan	Schaefer-Porter	Medicaid
aaRon MARTiNeZ	38	Female	A-	Hypertension	13-08-2023	Douglas Mayo	Lyons-Blair	Medicaid
connOR HANsEn	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletcher	Powers Miller, and Flores	Cigna
rObert bAuer	68	Female	AB+	Asthma	22-05-2020	Theresa Freeman	Rivera-Gutierrez	UnitedHealth
bROOkE brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stewart	Morris-Arellano	UnitedHealth
MS. nAtalie gAMble	46	Female	AB-	Obesity	01-01-2023	Maria Dougherty	Cline-Williams	Blue Cross
haley perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer	Cervantes-Wells	UnitedHealth
mRS. JamiE cAMPBELl	38	Male	AB-	Obesity	08-03-2020	Justin Kim	Torres, and Harrison Jones	Cigna
LuKE BuRgEss	34	Female	A-	Hypertension	04-03-2021	Justin Moore Jr.	Houston PLC	Blue Cross
dANIEL schmidt	63	Male	B+	Asthma	15-11-2022	Denise Galloway	Hammond Ltd	Cigna

Buttons: Extract Table Using Examples | Load | Transform Data | Cancel

Age Group Creation

The patient's age was converted into meaningful demographic groups to make analysis easier.

The age categories used were:

- 0–18
- 19–35
- 36–60
- 60 above

The Age Group column allows patients to be analyzed based on demographic ranges instead of individual ages.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Age Group

	Column Name	Operator	Value	Output
If	Age	is less than or equ...	18	0-18
Else If	Age	is less than or equ...	35	19-35
Else If	Age	is less than or equ...	60	36-60

Add Clause

Else

60 above

OK

Cancel

Standardizing Admission Type

The **Admission Type** field was standardized into meaningful categories:

- Emergency
- Elective
- Urgent

This ensures consistency when analyzing different types of hospital admissions.

Table.AddIndexColumn("#Removed Columns", "Index", 0, 1, Int64.Type)										
1.2	Billing Amount	1.3	Room Number	Admission Type	Discharge Date	Medication	Test Results	Age Group		
	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%	Valid Error Empty 100% 0% 0%		
1	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal	19-35			
2	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive	60 above			
3	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal	60 above			
4	37909.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal	19-35			
5	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal	36-60			
6	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal	36-60			
7	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive	19-35			
8	45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive	19-35			
9	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal	60 above			
10	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive	36-60			
11	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal	60 above			
12	7999.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive	36-60			
13	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal	60 above			
14	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal	60 above			
15	40701.59923	182	Urgent	13-10-2021	Paracetamol	Normal	36-60			
16	12263.35743	465	Elective	11-01-2023	Aspirin	Inconclusive	36-60			
17	24499.8479	114	Elective	14-07-2020	Paracetamol	Normal	60 above			
18	17440.46544	449	Urgent	02-04-2020	Paracetamol	Abnormal	36-60			
19	18843.02302	260	Elective	14-03-2021	Aspirin	Abnormal	19-35			
20	23762.20358	465	Elective	22-11-2022	Penicillin	Normal	60 above			
21	42.51458855	115	Elective	02-07-2023	Aspirin	Normal	60 above			
22	17695.91162	295	Urgent	09-02-2020	Lipitor	Normal	36-60			
23	5998.102908	327	Urgent	10-06-2022	Lipitor	Inconclusive	36-60			
24	25250.05243	119	Urgent	12-08-2021	Lipitor	Inconclusive	36-60			
25	33211.29542	109	Emergency	08-06-2020	Paracetamol	Inconclusive	60 above			
26										

Merge Queries

One of the important Power Query steps in the Week 5 project was **Merge Queries**.

Purpose of Merge

The Merge operation was used to combine related information from different tables based on a common key.

The common identifiers, such as **Patient Key** and **Admission ID/Admission Key**, were used to connect the appropriate records.

This allowed the required patient, admission, and billing information to be brought together for further modeling.

Merge

Select a table and matching columns to create a merged table.

Fact_Billing



Name	1	Age	2	Gender	4	Blood Type	5	Medical Condition	Room Number	Admission Type	Discharge
Bobby Jackson		30		Male		B-		Cancer	328	Urgent	02-
Leslie Terry		62		Male		A+		Obesity	265	Emergency	26-
Danny Smith		76		Female		A-		Obesity	205	Emergency	07-
Andrew Watts		28		Female		O+		Diabetes	450	Elective	18-

Dim_Patient



Name	1	patient ID	Age	2	Age Group	3	Gender	4	Blood Type	5
Bobby Jackson		1	30		19-35		Male		B-	
Leslie Terry		2	62		60above		Male		A+	
Danny Smith		3	76		60above		Female		A-	
Andrew Watts		4	28		19-35		Female		O+	
Adrienne Bell		5	43		36-60		Female		Ab+	

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

► Fuzzy matching options

✓ The selection matches 40666 of 55500 rows from the first table.

OK

Cancel

Result of the Merge

The Merge operation helped combine the required information while maintaining the relationship between patient, admission and billing records.

This transformed the source data into a structure that could be separated into:

- Dim_Patient
- Dim_Admission
- Fact_Billing

The merge step was therefore an important part of preparing the final data model.

Queries [5] Table.SelectColumns(Source, ("PatientID", "Name", "Age", "Age Group", "Gender", "Blood Type"))

PatientID	Name	Age	Age Group	Gender	Blood Type
1	Bobby Jackson	30	19-35	Male	B-
2	Leslie Terry	62	60 above	Male	A+
3	Danny Smith	76	60 above	Female	A-
4	Andrew Watts	28	19-35	Female	O+
5	Adrienne Bell	43	36-60	Female	AB+
6	Emily Johnson	36	36-60	Male	A+
7	Edward Edwards	21	19-35	Female	AB-
8	Christina Martinez	20	19-35	Female	A+
9	Jasmine Aguilar	82	60 above	Male	AB+
10	Christopher Berg	58	36-60	Female	AB-
11	Michelle Daniels	72	60 above	Male	O+
12	Aaron Martinez	38	36-60	Female	A-
13	Connor Hansen	75	60 above	Female	A+
14	Robert Bauer	68	60 above	Female	AB+
15	Brooke Brady	44	36-60	Female	AB+
16	Ms. Natalie Gamble	46	36-60	Female	AB-
17	Hailey Perkins	63	60 above	Female	A+
18	Mrs. Jamie Campbell	38	36-60	Male	AB-
19	Luke Burgess	34	19-35	Female	A-
20	Daniel Schmidt	63	60 above	Male	B+
21	Timothy Burns	67	60 above	Female	A-
22	Christopher Bright	48	36-60	Male	B+
23	Kathryn Stewart	58	36-60	Female	O+
24	Dr. Eileen Thompson	59	36-60	Male	A+
25	Paul Henderson	72	60 above	Female	AB+
26	Peter Fitzgerald	73	60 above	Male	AB+

Query Settings

PROPERTIES

Name: Dim_Patient

APPLIED STEPS

Source: Removed Other Columns

Queries [5] Table.ReorderColumns(*Renamed Columns, ("AdmissionID", "Medical Condition", "Hospital", "Doctor", "Medication", "Room Number", "Test Res"))

AdmissionID	Medical Condition	Hospital	Doctor	Medication	Room Number	Test Res
1	Cancer	Sons and Miller	Matthew Smith	Paracetamol	328	Normal
2	Obesity	Kim Inc	Samantha Davies	Ibuprofen	265	Inconclus
3	Obesity	Cook PLC	Tiffany Mitchell	Aspirin	205	Normal
4	Diabetes	Hernandez Rogers and Vang,	Kevin Wells	Ibuprofen	450	Abnorma
5	Cancer	White-White	Kathleen Hanna	Penicillin	458	Abnorma
6	Asthma	Nunez-Humphrey	Taylor Newton	Ibuprofen	389	Normal
7	Diabetes	Group Middleton	Kelly Olson	Paracetamol	389	Inconclus
8	Cancer	Powell Robinson and Valdez,	Suzanne Thomas	Paracetamol	277	Inconclus
9	Asthma	Sons Rich and	Daniel Ferguson	Aspirin	316	Abnorma
10	Cancer	Padilla-Walker	Heather Day	Paracetamol	249	Inconclus
11	Cancer	Schaefer-Porter	John Duncan	Paracetamol	394	Normal
12	Hypertension	Lyons-Blair	Douglas Mayo	Lipitor	288	Inconclus
13	Diabetes	Powers Miller, and Flores	Kenneth Fletcher	Penicillin	134	Abnorma
14	Asthma	Rivera-Gutierrez	Theresa Freeman	Lipitor	309	Normal
15	Cancer	Morris-Arellano	Roberta Stewart	Paracetamol	182	Normal
16	Obesity	Cline-Williams	Maria Dougherty	Aspirin	465	Inconclus
17	Arthritis	Cervantes-Wells	Erica Spencer	Paracetamol	214	Normal
18	Obesity	Torres, and Harrison Jones	Justin Kim	Paracetamol	449	Abnorma
19	Hypertension	Houston PLC	Justin Moore Jr.	Aspirin	260	Abnorma
20	Asthma	Hammond Ltd	Denise Galloway	Penicillin	465	Normal
21	Asthma	Jones LLC	Krista Smith	Aspirin	115	Normal
22	Asthma	Williams-Davis	Gregory Smith	Lipitor	295	Normal
23	Arthritis	Clark-Mayo	Vanessa Newton	Lipitor	327	Inconclus
24	Asthma	and Sons Smith	Donna Martinez MD	Lipitor	119	Inconclus
25	Hypertension	Wilson Group	Stephanie Kramer	Paracetamol	109	Inconclus

Query Settings

PROPERTIES

Name: Dim_Admission

APPLIED STEPS

Source: Removed Other Columns, Removed Columns, Added Index, Renamed Columns, Reordered Columns

Queries [5] Table.RenameColumns(Added Index4,({"Index", "AdmissionKey"}, {"PatientKey", "PatientKey"}))

	Date of Admission	Insurance Provider	1.2 Billing Amount	Discharge Date	12 PatientKey	12 AdmissionKey
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	31-01-2024	Blue Cross	18856.28131	02-02-2024	1	1
2	20-08-2019	Medicare	33643.32729	26-08-2019	2	2
3	22-09-2022	Aetna	27955.09608	07-10-2022	3	3
4	18-11-2020	Medicare	37909.78241	18-12-2020	4	4
5	19-09-2022	Aetna	14238.31781	09-10-2022	5	5
6	19-09-2022	Aetna	14238.31781	09-10-2022	6	6
7	19-09-2022	Aetna	14238.31781	09-10-2022	7	7
8	19-09-2022	Aetna	14238.31781	09-10-2022	8	8
9	20-12-2023	UnitedHealthcare	48145.11095	24-12-2023	9	9
10	09-11-2020	Medicare	19580.87234	15-11-2020	10	10
11	28-12-2021	Cigna	45820.46272	07-01-2022	11	11
12	01-07-2020	Cigna	50119.22279	14-07-2020	12	12
13	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021	13	13
14	19-04-2020	Medicare	12576.79561	22-04-2020	14	14
15	13-08-2023	Medicare	7999.58688	05-09-2023	15	15
16	12-12-2019	Cigna	43282.28336	28-12-2019	16	16
17	22-05-2020	UnitedHealthcare	33207.70663	19-06-2020	17	17
18	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021	18	18
19	01-01-2023	Blue Cross	12263.35743	11-01-2023	19	19
20	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020	20	20
21	08-03-2020	Cigna	17440.46544	02-04-2020	21	21
22	04-03-2021	Blue Cross	18843.02302	14-03-2021	22	22
23	15-11-2022	Cigna	23762.20358	22-11-2022	23	23
24	28-06-2023	Blue Cross	42.51458855	02-07-2023	24	24
25	21-01-2020	Aetna	17695.91162	09-02-2020	25	25
26	12-05-2022	Aetna	5998.102908	10-06-2022	26	26

Query Settings

PROPERTIES

Name

Fact_Billing

APPLIED STEPS

Added Index1

Renamed Columns1

Added Custom

Removed Columns

Added Index2

Renamed Columns2

Changed Type1

Renamed Columns3

Reordered Columns

Removed Columns1

Removed Duplicates

Merged Queries

Expanded Dim_Patient

Merged Queries1

Expanded Dim_Admission

Removed Other Columns

Removed Columns2

Renamed Columns4

Removed Columns3

Added Index3

Renamed Columns5

Removed Columns4

Added Index4

Renamed Columns6

Data Modeling

After cleaning the data, the dataset was organized into a structured model using a **star-schema approach**.

The main tables created were:

Dimension Tables

Dim_Patient

The patient dimension contains patient-related information such as:

- Patient ID
- Name
- Age
- Age Group
- Gender
- Blood Type

Dim_Admission

The admission dimension contains admission and medical-related information such as:

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication

- Room Number
- Test Results

Fact Table

Fact_Billing

The fact table contains transactional/admission-related information required for healthcare and billing analysis, including:

- Admission Key
- Patient Key
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider

Measures Table

A separate **Measures Table** was created to organize the DAX measures used in the report.

Star Schema Relationships

The final model follows a star-schema structure where **Fact_Billing** acts as the central fact table.

The main relationships are:

Dim_Admission → Fact_Billing and **Dim_Patient → Fact_Billing**

The model uses a **many-to-one (*:1) relationship**, where one patient or admission record can be associated with multiple records in the fact table.

The healthcare_dataset table was retained as the cleaned/source dataset used during the transformation and modeling process.

DAX Measures Created

Several DAX measures were created to calculate important healthcare KPIs.

1. Total Patients / Admissions

Total Patients = COUNTROWS(Fact_Billing)

This measure counts the number of records in the Fact_Billing table and is used to represent the total number of billing/admission records in the model.

2. Total Healthcare Revenue

Total Billing Amount =

SUM(Fact_Billing[Billing Amount])

This measure calculates the total billing amount across all records.

It can be used to understand the overall healthcare revenue represented in the dataset.

3. Average Billing Cost per Patient

AVG Billing Amount =

AVERAGE(Fact_Billing[Billing Amount])

This measure calculates the average billing amount across the records in Fact_Billing.

It provides an overview of the typical billing amount associated with an admission record.

4. Average Length of Stay

AVG Length of Stay =

AVERAGEX(
Fact_Billing,
DATEDIFF(
Fact_Billing[Date of Admission],
Fact_Billing[Discharge Date],
DAY
)
)

This measure calculates the number of days between the admission date and discharge date for each record and then calculates the average.

It can be used to analyze the typical duration of a patient's hospital stay.

5. Readmission Rate

The Readmission Rate measure was created using the patient's PatientKey and the number of admission records associated with each patient.

The measure identifies patients who have more than one admission and calculates their proportion relative to the total patient/admission count.

The measure structure used was:

Readmission Rate =

VAR TotalPatients =

DISTINCTCOUNT(Fact_Billing[PatientKey])

VAR RepeatPatients =

```

COUNTROWS(
    FILTER(
        SUMMARIZE(
            Fact_Billing,
            Fact_Billing[PatientKey],
            "AdmissionCount", COUNTROWS(Fact_Billing)
        ),
        [AdmissionCount] > 1
    )
)

RETURN
    DIVIDE(RepeatPatients, TotalPatients, 0)

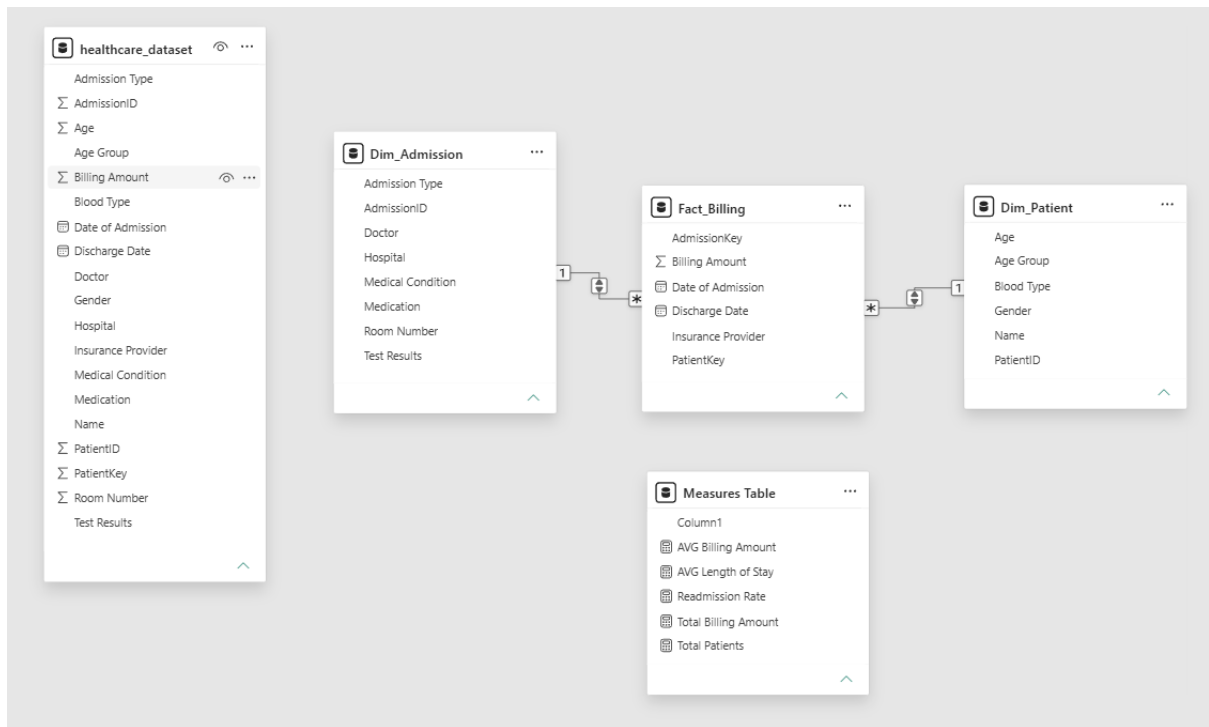
```

Final Data Model

The completed model contains the following major components:

- **healthcare_dataset** – cleaned/source healthcare dataset
- **Dim_Patient** – patient-related dimension
- **Dim_Admission** – admission-related dimension
- **Fact_Billing** – central fact table containing billing and admission-level information
- **Measures Table** – stores the DAX measures used for analysis

The model provides a structured foundation for analyzing healthcare information through Power BI.



DAX Measures Table

The measures were organized in a separate **Measures Table**, making the Power BI model cleaner and easier to manage.

The completed Measures Table contains:

- AVG Billing Amount
- AVG Length of Stay
- Readmission Rate
- Total Billing Amount
- Total Patients

Measures Table	
Column1	
AVG Billing Amount	
AVG Length of Stay	
Readmission Rate	
Total Billing Amount	
Total Patients	

Conclusion

The Week 5 Healthcare Power BI project provided practical experience in **data cleaning, transformation, dimensional modeling, relationship creation and DAX calculations**.

By separating patient and admission information into dimension tables and maintaining billing/admission records in the fact table, the healthcare data was organized into a structured analytical model.

The created KPIs, including **Total Patients, Total Billing Amount, Average Billing Amount, Average Length of Stay, and Readmission Rate**, provide a foundation for analyzing patient admissions and healthcare billing information.

Overall, the completed model is **clean, structured and ready for healthcare data analysis in Power BI**.